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RESEARCH ARTICLE

Safety Culture Transformation in Mega Projects: The Role of Leadership in Sustainable Performance

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ABSTRACT

Megaprojects are known for their budgets exceeding one billion dollars, prolonged timelines, and a complex contractor's management environment. They are at the center of industrial development, specifically in the Gulf Cooperation Council (GCC) region under initiatives such as the United Arab Emirates' sustainability agenda and Saudi Vision 2030. Despite advances in technical controls and risk systems, megaprojects still struggle to prevent serious injuries and fatalities. This indicates that compliance alone does not create a mature safety culture. This paper reviews the literature on safety culture transformation, leadership accountability, and mega-project governance, generating lessons from global case studies such as the Macondo/Deepwater Horizon disaster, the United Kingdom's cross-rail project, and Qatar's World Cup construction programs. The analysis demonstrates that leadership accountability at board, project director, and site levels is an essential element for cultural maturity, shifting organizations from dependent or independent safety stages to interdependent, high-performing cultures. In this paper, a three-level accountability cascade framework is proposed to link executive governance with frontline behaviors. Implications are discussed for GCC megaprojects, where reputational, ESG, and legal pressures come together. The paper concludes that without executive-level ownership, safety culture transformation in megaprojects is becoming superficial.

Keywords: Leadership accountability, mega-projects, safety culture, safety transformation.

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1. Introduction

Megaprojects have evolved as engines of national development and focal points of occupational health and safety (OHS) and risk. Large-scale projects exceeding USD 1 billion in cost usually last for multiple years and deal with complex contractor ecosystems. Megaprojects are established in sectors such as oil and gas, infrastructure, mining, and renewable energy (Flyvbjerg, 2017). In the Gulf Cooperation Council (GCC) region, leading initiatives such as Saudi Arabia's Vision 2030 giga-projects (e.g., NEOM and the Red Sea Project), the UAE's post-Expo 2020, and in Oman, the Rabab Harweel Integrated Project (RHIP) highlight the difficult safety and leadership challenges faced by GCC mega-projects.

Despite their visibility, mega-projects are consistently associated with cost overruns, schedule delays, and safety incidents (Cantarelli *et al*., 2012; Merrow, 2011).

From an HSE perspective, mega-projects present unique challenges, such as multi-contractor and subcontractor hierarchies, multinational workforces with different safety norms, harsh environmental conditions, and elevated production pressures (Antonsen, 2017). As Reason (1997) and Hudson (2007) show, long-term safety performance depends less on technical systems and more on the maturity of the shared values and practices. However, safety culture transformation in mega-projects is often superficial, with initiatives focusing on compliance or visible campaigns rather than driving deep behavioral change (Cooper, 2000; Guldenmund, 2010).

Simultaneously, leadership accountability has become central to global safety discussions. The United Kingdom's Corporate Manslaughter and Corporate Homicide Act (2007), the European Union's Corporate Sustainability Due Diligence Directive (proposed in 2022), and the

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International Council on Mining and Metals (ICMM) leadership principles emphasize executives' personal and institutional accountability for safety outcomes (Hopkins, 2014; Dekker, 2012; International Council on Mining & Metals (ICMM), 2020). This accountability is more than a legal liability; it is progressively linked to ESG ratings, investor confidence, and stakeholder legitimacy (Bice & Moffat, 2014).

This study explores the intersection of safety culture transformation, leadership accountability, and mega-project dynamics. It argues that leadership accountability is the missing multiplier in enabling megaprojects to move beyond symbolic compliance to interdependent, sustainable safety cultures. Building on literature, global case studies, and theoretical frameworks, this study proposes a three-level accountability (strategic, tactical, andoperational) as a practical model for integrating accountability across the lifecycle of mega-projects.

2. Literature Review

2.1. Mega-Projects Safety Challenges

Mega projects are complex sociotechnical projects (Flyvbjerg, 2017). These characteristics include long development cycles, political visibility, and global supply chains. Within this landscape, safety is a complex challenge: risks are systemic, failures can spread quickly, and accountability is often unclear (Merrow, 2011).

Research highlights four recurring safety challenges in megaprojects: (1) contractor fragmentation, (2) multinational workforces, (3) production pressures, and (4) harsh environments. Multiple levels of subcontractors often result in weak oversight and diluted accountability; studies on construction mega-projects in the Middle East reveal that subcontractors frequently operate under cost and schedule pressures that encourage unsafe practices (Alkilani *et al*., 2013). Mega-projects hire labor from diverse countries with varying languages, cultures, and risk perceptions, complicating the establishment of a common safety culture (Fleming & Lardner, 2002). High-profile deadlines encourage risk-taking behaviors, leading to accidents and reputational crises, as highlighted by the Qatar 2022 construction program (Amnesty International, 2021). In the GCC region, extreme heat and climatic conditions introduce unique occupational hazards and require adaptive safety strategies (Khalfan *et al*., 2020).

2.2. Safety Culture Models

The concept of safety culture began in the wake of the Chernobyl disaster (International Atomic Energy Agency (IAEA), 1986) and became the center of OHS research and practice. Several models frame cultural maturity and organizational learning. Reason's (1997) Swiss Cheese Model conceptualizes accidents as alignments of latent organizational failures with active errors, emphasizing systemic causation. Hudson's (2007) Safety Culture Ladder describes the progression from pathological to generative cultures and is widely used in oil and gas. The DuPont Bradley Curve maps stage-reactive, dependent, independent, and interdependent processes on the journey from compliance to shared accountability (DuPont, 1994). UK HSE maturity models emphasize leadership, worker engagement, and continuous improvement.

These frameworks underline the transition from compliance-driven behaviors to ownership and interdependence. However, critics argue that many organizations use them as branding exercises, claiming maturity without embedding behaviors (Cooper, 2000; Guldenmund, 2010). This critique is especially significant in megaprojects, where visible campaigns can suppress persistent structural weaknesses.

2.3. Leadership Accountability in Safety

Leadership has consistently been identified as the strongest predictor of safety culture outcomes (Zohar, 2010). The question is not whether leadership matters but which dimensions of leadership—symbolic, operational, and accountable—drive transformation in complex, multi-actor projects.

2.4. Regulatory and Legal Accountability

The UK Corporate Manslaughter and Corporate Homicide Act (2007) established a standard for holding organizations criminally liable for systemic safety failures. In the European Union, the proposed Corporate Sustainability Due Diligence Directive (European Commission, 2022) requires directors to oversee human rights and environmental risks, thereby expanding the scope of accountability into ESG domains. In mining and energy, ICMM leadership principles emphasize visible safety commitments and integration into governance (International Council on Mining and Metals (ICMM), 2020). These frameworks place more responsibility on boards and senior executives than, just supervisors.

2.5. Leadership Behaviors Linked to Culture

Empirical studies suggest that leadership shapes safety culture through four mechanisms: (a) vision and prioritization—signaling in which safety is equal to production (Clarke, 2013); (b) resource allocation—funding for welfare, PPE, and training (Hale *et al*., 2010); (c) learning orientation—leaders who respond to incidents with systemic inquiry foster psychological safety and reporting (Edmondson, 2018); and (d) governance integration—linking safety outcomes to board-level ESG reporting and incentive structures (Bice & Moffat, 2014).

2.6. The Accountability Gap in Mega-Projects

In megaprojects, responsibility often dilutes as boards delegate to project directors, who delegate to contractors, leading to a cascade of abdication (Hopkins, 2014). When incidents occur, accountability is frequently assigned to the lowest level, while strategic decisions remain unexamined. This accountability gap undermines the transformation of safety culture and sustains a blame culture.

2.7. Case Studies: Lessons from Global Mega-Projects

National Commission on the BP Deepwater Horizon Oil Spill and Offshore Drilling (2011) investigated leadership

failures at BP, Transocean, and Halliburton, including cost pressures that overriding safety concerns. This case revealed how fragmented accountability can undermine safety in multifactor projects.

Crossrail, United Kingdom: One of Europe's largest infrastructure projects, Crossrail invested heavily in safety culture programs, embedding leadership accountability through KPIs and board oversight (Crossrail Learning Legacy, 2018). It achieved significant improvements but still faced challenges in contractor alignment.

Qatar 2022 World Cup Construction: International criticism highlighted poor labor conditions and safety incidents. Reforms, including revisions to the Kafala system, were driven not only by local enforcement but also by reputational accountability to global stakeholders (Amnesty International, 2021).

NEOM, Saudi Arabia (ongoing): While data is still emerging, NEOM positions safety and sustainability as brand elements. Embedding leadership accountability in giga-project governance is decisive for legitimacy.

3. Linking Leadership Accountability to Safety Culture Transformation

3.1. The Accountability Gap

Safety culture efforts in megaprojects often stall because accountability is fragmented. Boards and senior executives issue safety policies; however, day-to-day decisions are delegated to project managers and contractors. The ownership of outcomes often stops at middle management, leading to responsibility being passed down and avoided (Hopkins, 2014). This gap manifests as (1) strategic–operational disconnect, (2) persistence of blame culture, and (3) surface-level gestures such as checklist walkarounds.

3.2. Mechanisms by Which Leadership Shapes Culture

- *Tone from the Top:* Leadership statements and behaviors significantly influence worker perceptions of safety priorities (Clarke, 2013). If production bonuses outweigh safety metrics, workers perceive that speed matters more than safety do.
- *Resource Allocation:* Investment in welfare measures (e.g., shaded rest, hydration stations, climate-appropriate PPE) signals leadership accountability and reduces risk in harsh environments (Hale *et al*., 2010; Khalfan *et al*., 2020).
- *Learning Orientation:* Leaders who respond to incidents with systemic inquiry rather than punishment foster psychological safety and encourage reporting (Edmondson, 2018; Dekker, 2012).
- *Governance Integration:* Linking safety outcomes into board-level ESG reporting and remuneration closes the accountability gap and keeps attention on safety beyond campaigns (Bice & Moffat, 2014; International Council on Mining & Metals (ICMM), 2020).

3.3. Case Insights

Deepwater Horizon (2010) demonstrated how executive cost-cutting decisions cascaded into operational risk; leadership failed to integrate safety metrics into strategic decision making (National Commission on the BP Deepwater Horizon Oil Spill and Offshore Drilling, 2011). Crossrail showed how visible leadership, mandatory safety KPIs, and contractor alignment created a more consistent culture across a fragmented workforce (Crossrail Learning Legacy, 2018). Qatar 2022 revealed the force of reputational accountability in which global stakeholders drove reform. Together, these insights imply that leadership accountability must be legal, reputational, operational and cultural.

4. Proposed Accountability Cascade Framework

4.1. Rationale

Existing safety culture models provide maturity stages but minimize executive governance. Governance frameworks emphasize legal and reputational accountability but often miss the link to daily worker behavior. The Three-Level Accountability Cascade Framework integrates both perspectives to operationalize leadership accountability across strategy, program management, and frontline practice.

4.2. Framework Overview

- *Level 1–Strategic (Board & C-Suite):* Embed safety into corporate governance, ESG reporting, and financial decision-making. Actions include linking executive compensation to safety KPIs, integrating safety into investment gate reviews and transparent public reporting. For example, some ICMM members link safety metrics to executive bonuses (International Council on Mining & Metals (ICMM), 2020).
- *Level 2–Tactical (Project Directors & Senior Managers):* Align contractors and supply chains with safety culture expectations. Actions include safety clauses in procurement contracts, joint safety leadership programs, and cascading KPIs across subcontractors.
- *Level 3–Operational (Supervisors & Workers):* Translate leadership commitment into daily practices via supervisor safety conversations, peer-to-peer accountability, near-miss reporting systems, and stopwork authority. ExxonMobil's Operations Integrity Management System (OIMS) is a reference for frontline empowerment (ExxonMobil, 2019).

The cascading structure is illustrated in Fig. 1.

4.3. Benefits of the Framework

The framework offers several benefits: (1) closing the accountability gap so that responsibility does not stall in middle management; (2) aligning culture with governance to bridge campaigns and board-level consequences; (3) scaling a consistent accountability structure across

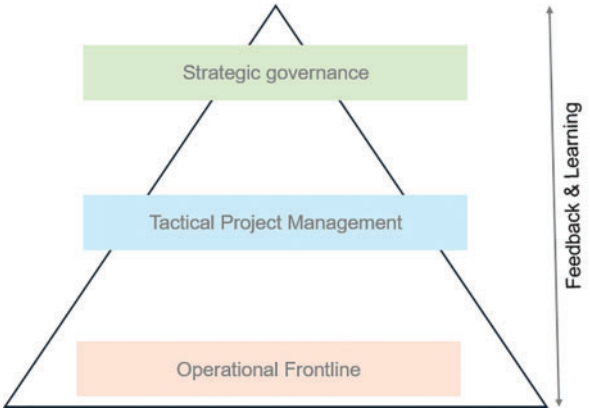


Fig. 1. Cascading triangle framework for safety culture transformation.

multinational workforces; and (4) integrating safety into sustainability reporting demanded by global investors.

5. Discussion

Megaprojects present a paradox: while they are highly resourced and strategically important, they frequently underperform on safety compared to smaller projects (Flyvbjerg, 2017; Merrow, 2011). This study argues that the key missing factor is leadership accountability, which decides whether safety culture stays aspirational or becomes embedded.

5.1. Implications for GCC Mega-Projects

The GCC context illustrates these dynamics in detail: Saudi Arabia's Vision 2030 giga-projects (e.g., NEOM, the Red Sea Project) and the UAE's energy and infrastructure expansions involve multinational workforces operating under harsh climatic conditions and geopolitical scrutiny. In such contexts, reputational risk is global, and safety incidents attract media and NGO attention, as seen in Qatar 2022s'construction projects (Amnesty International, 2021). Investor pressure is rising, with ESG criteria now explicitly including worker safety and well-being (Bice & Moffat, 2014). Regulatory frameworks are tightening as GCC countries align with international OHS standards, although enforcement varies (Khalfan *et al*., 2020). Embedding leadership accountability through the proposed cascade framework ensures that strategic ambitions (e.g., sustainability, ESG) are operationalized into daily practices.

5.2. Risks and Challenges

Implementation faces obstacles such as symbolic compliance (polished reports without substantive change) (Cooper, 2000); cultural resistance in hierarchical settings that discourage upward feedback (Antonsen, 2017); and data overload in digital dashboards that create vanity metrics unless tied to meaningful interventions (Hopkins, 2019).

5.3. Future Directions

Future research should explore (1) digital accountability tools—how AI-powered dashboards and predictive analytics can strengthen, rather than dilute, leadership accountability; (2) cross-cultural safety leadership, how multinational workforces perceive accountability signals across cultural contexts; and (3) longitudinal evidence, tracking how accountability reforms influence safety culture maturity over the lifecycle of mega-projects.

6. Conclusion

Megaprojects carry the highest stakes in safety: huge investments, complex organizations, and serious risks of catastrophic failure. Safety culture transformation cannot be achieved through technical systems or frontline initiatives. The critical multiplier is leadership accountability embedded at strategic, tactical, and operational levels. The proposed Three-Level Accountability Cascade Framework offers a practical model to align governance with culture. By integrating board-level oversight, project-level management, and frontline ownership, mega-projects can move beyond compliance-driven safety and toward a culture built on shared responsibility and resilience. For GCC megaprojects, this approach is particularly urgent. With Vision 2030 and other national strategies drawing global attention to the GCC, leadership accountability has become a reputational, financial, and ethical necessity.

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Conflict of Interest

The authors declare that they do not have any conflict of interest.

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